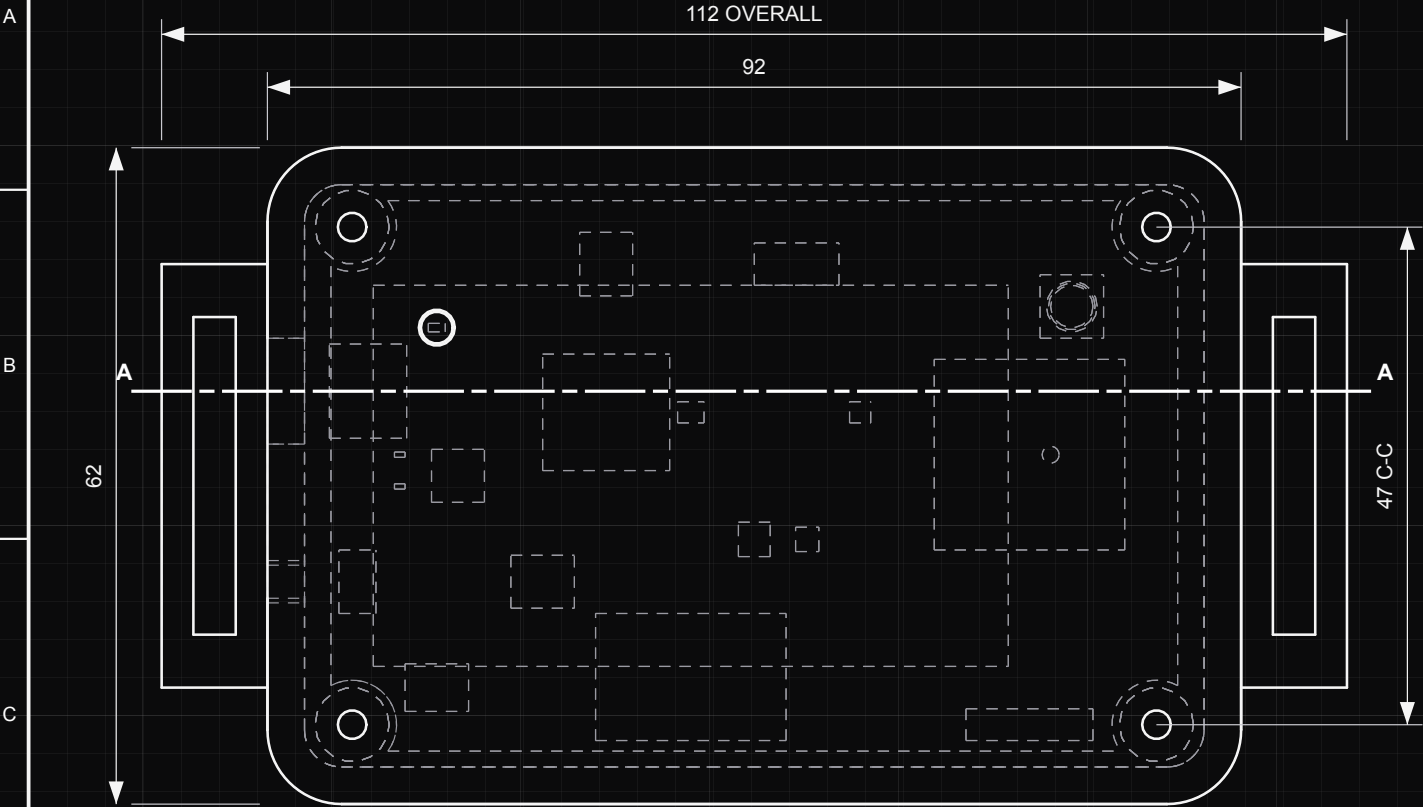
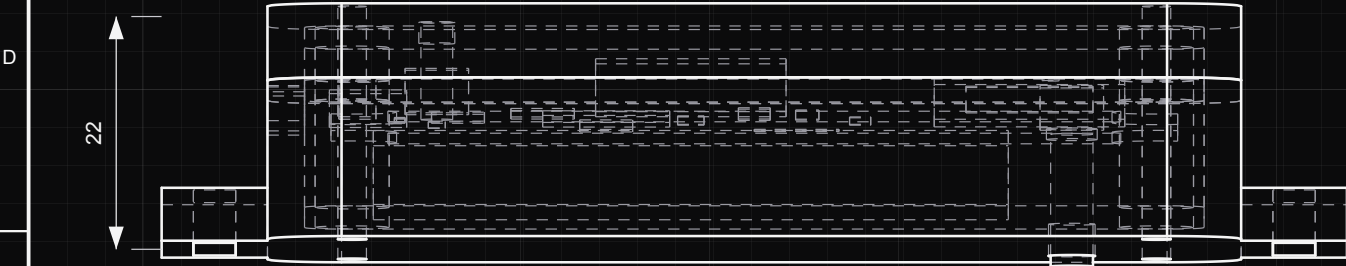


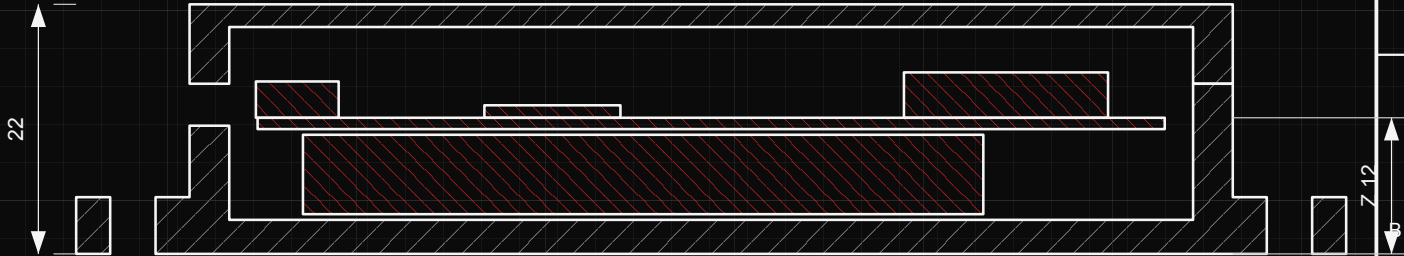
REV	DESCRIPTION	DATE
0	Hardware concept RL-MB-01	2026-10-05



PLAN VIEW (LID SHOWN TRANSPARENT) / 1.4:1

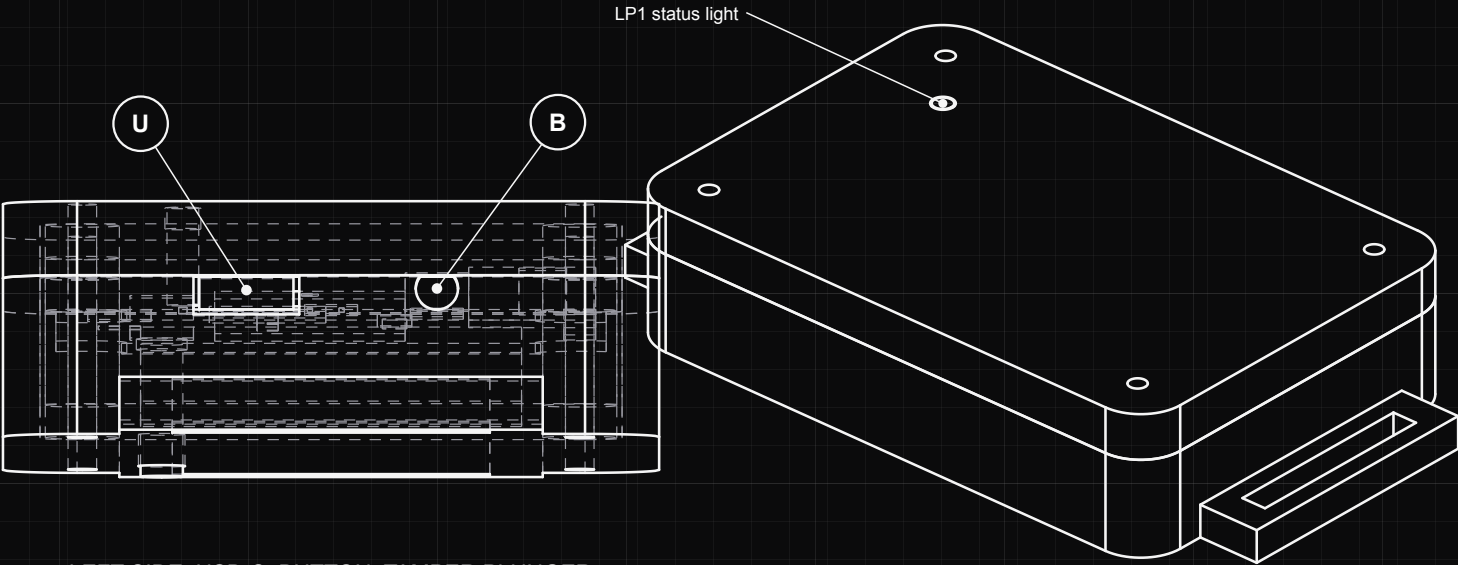


FRONT ELEVATION



SECTION A-A / Y = +8 / 1.5:1

Enclosure white hatch, electronics red hatch (PCB1, U1, U7, J1, BT1)



ISOMETRIC, ASSEMBLED / 1:1

LEFT SIDE: USB-C, BUTTON, TAMPER PLUNGER

U USB-C opening 10 x 4.6 (charge only), silicone plug

B hole D4 for SW1, membrane to be added

GENERAL NOTES

- Dimensions in mm. Third-angle projection. General tolerances to be defined (ISO 2768-m proposed).
- Board outlines: maker or distributor data cited in hardware/netlist.json; others are indicative envelopes.
- Heights, connectors and positions of openings are indicative and must be measured on the chosen parts.
- Geometry and sheets generated by cad/generate.py (build123d 0.13.0); checks run with --check.

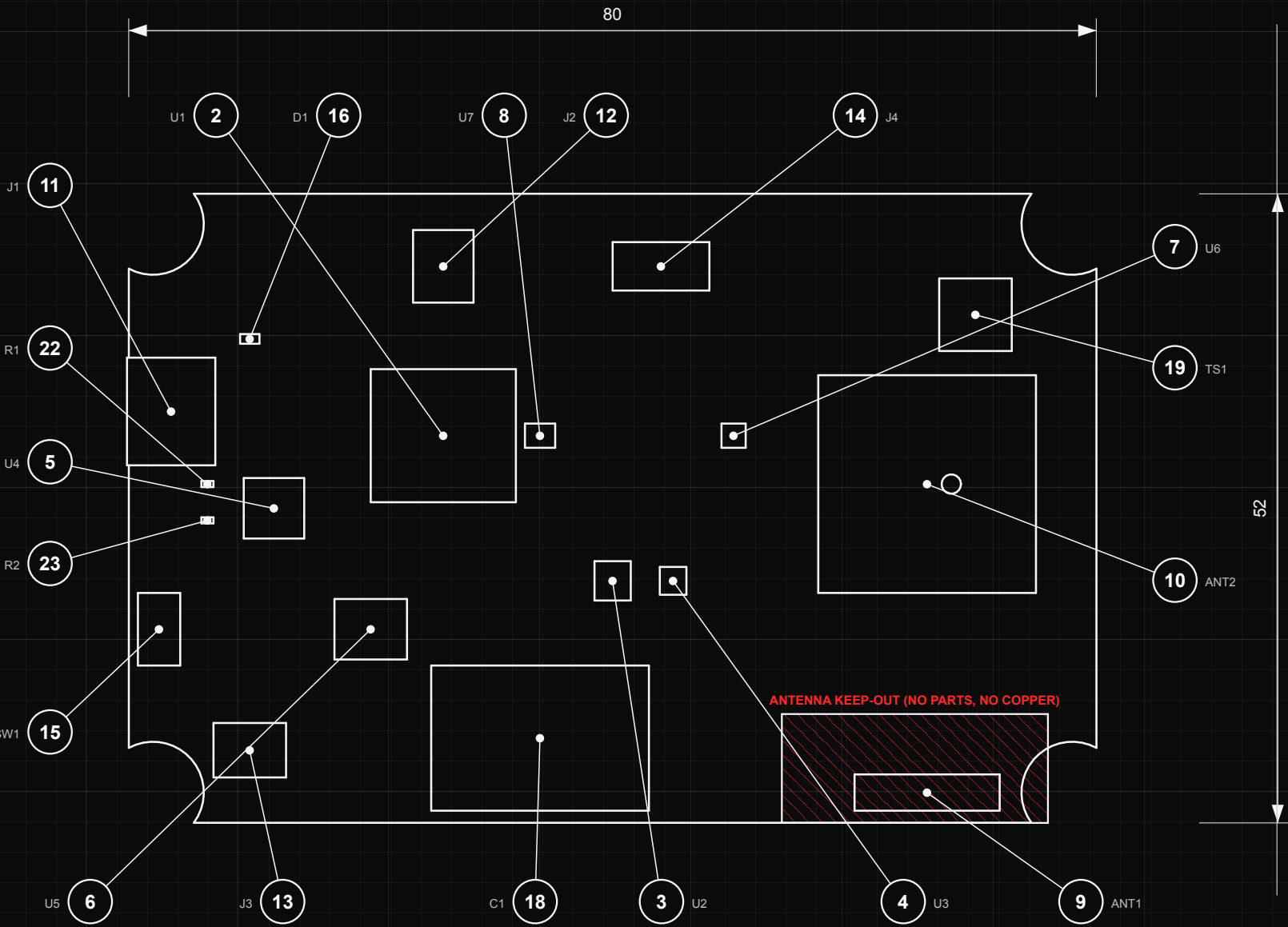
GENERAL ASSEMBLY

Redline tracker and black box, board RL-MB-01 (nRF9151)

DRAWING NO. RL-HW-01	SCALE 1.4:1 unless noted	REV 0	SHEET 1 / 5
DRAWN BY: M. ABAAQIL	DATE: 2026-10-05	MATERIAL: to be defined	

REDLINE / RL-MB-01

CONCEPT STUDY - PROVISIONAL DIMENSIONS - DO NOT BUILD WITHOUT REVIEW



01 BOARD RL-MB-01, COMPONENT SIDE / 2:1

Thickness 1 - notches R4.2 at the four screws - top face at Z = 12

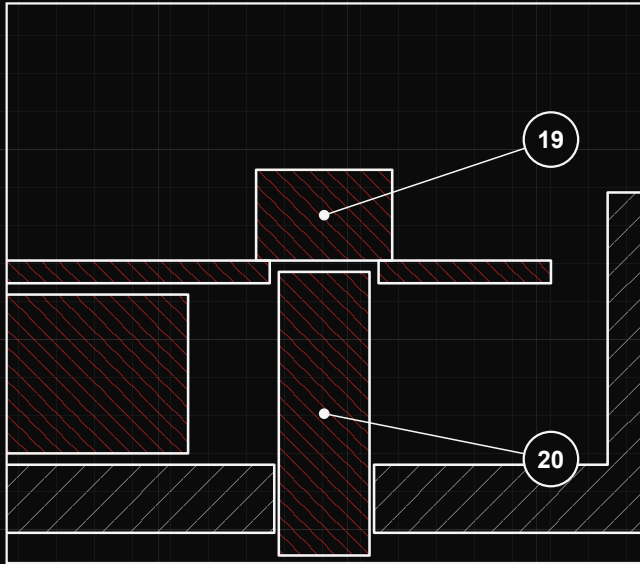
LAYOUT RULES

- ANT1 at the board edge, keep-out clear of copper and parts (checked).
- ANT2 faces the polymer lid; nothing metallic above it.
- U2 near the centre of the board, held by four rigid points.
- BT1 under the board: check the antenna tuning with it in place.
- J1 and SW1 line up with the openings in the -X wall (checked).

GENERAL NOTES

- Dimensions in mm. Third-angle projection. General tolerances to be defined (ISO 2768-m proposed).
- Board outlines: maker or distributor data cited in hardware/netlist.json; others are indicative envelopes.
- Heights, connectors and positions of openings are indicative and must be measured on the chosen parts.
- Geometry and sheets generated by cad/generate.py (build123d 0.13.0); checks run with --check.

REV	DESCRIPTION	DATE
0	Hardware concept RL-MB-01	2026-10-05

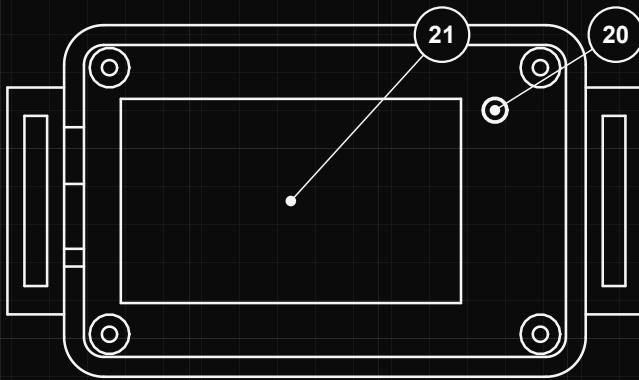


DETAIL B: TAMPER PLUNGER, SECTION Y = +16 / 3:1

Mounting surface pushes PL1 up: TS1 stays open. Remove the device: spring drops PL1, TS1 closes.



LID, INSIDE FACE / 0.75:1: D3.4 hole for the light pipe, no metal over ANT2

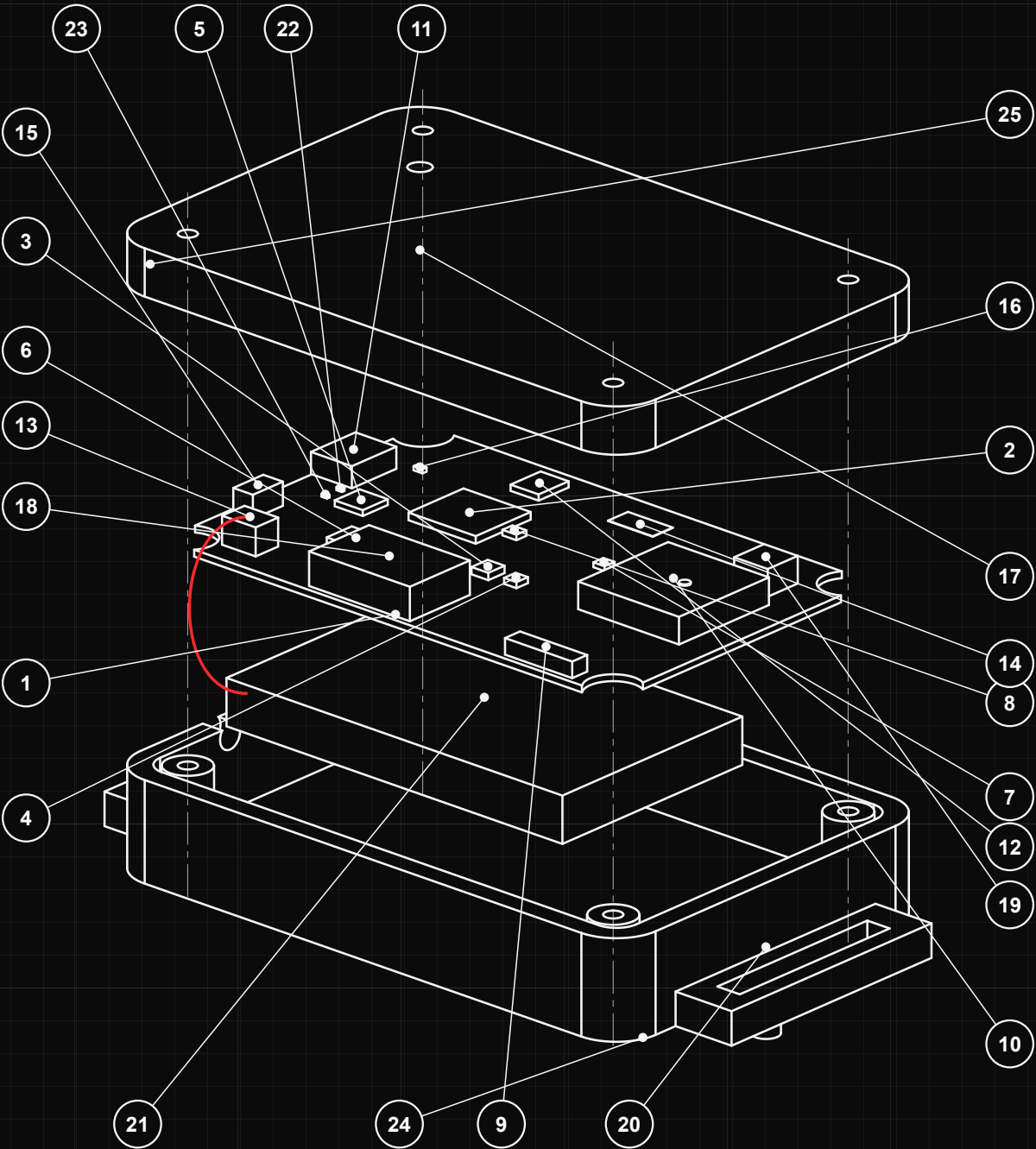


BASE WITH BT1 ON FOAM AND PL1 (BOARD REMOVED) / 0.75:1

BOARD RL-MB-01 AND TAMPER MECHANISM

Redline tracker and black box, board RL-MB-01 (nRF9151)

DRAWING NO. RL-HW-02	SCALE 2:1 board, 3:1 detail	REV 0	SHEET 2 / 5
DRAWN BY: M. ABAAQIL	DATE: 2026-10-05	MATERIAL: to be defined	



EXPLODED ISOMETRIC / 25 CALLOUTS / SCREW AXES IN CHAIN LINE / W1 BATTERY CABLE IN RED

PARTS LIST

REV	DESCRIPTION	DATE
0	Hardware concept RL-MB-01	2026-10-05

NO	QTY	DESIGNATION	PART	SIZE mm	DATA
1	1	PCB1 Main board RL-MB-01	Redline design	80 x 52	design choice
2	1	U1 nRF9151 SiP- LTE-M, NB-IoT, NTN, DECT NR+, GNSS	Nordic nRF9151	12 x 11	maker (height indicative)
3	1	U2 ADXL372 shock sensor, +-200 g	Analog Devices ADXL372	3 x 3.25	maker
4	1	U3 ADXL367 motion sensor, +-2/4/8 g	Analog Devices ADXL367	2.2 x 2.3	maker
5	1	U4 nPM1300 PMIC: charger, fuel gauge, regulators	Nordic nPM1300 QFN32	5 x 5	maker (height indicative)
6	1	U5 64 Mbit low-power flash: the black box memory	Macronix MX25R6435F	6 x 5	indicative (WSON-8)
7	1	U6 GNSS filter then LNA	to choose	2 x 2	indicative
8	1	U7 Hold-up charger and ideal diode for C1	to choose	2.5 x 2	indicative
9	1	ANT1 Chip antenna 824-960 / 1710-2690 MHz	Ignion NN02-224	12 x 3	distributor
10	1	ANT2 GNSS patch antenna 18 mm	Taoglas CGGBP.18.4.A.02	18 x 18	maker
11	1	J1 USB-C receptacle (charge and black box readout)	to choose	7.3 x 8.9	indicative
12	1	J2 eSIM, soldered MFF2	to choose (IoT operator)	5 x 6	indicative
13	1	J3 Battery connector JST-PH 2, SMD	to choose	6 x 4.5	indicative
14	1	J4 SWD pads Tag-Connect TC2030	footprint only	8 x 4	footprint
15	1	SW1 Side button: arm, disarm, pair	to choose	3.5 x 6	indicative
16	1	D1 Status LED 0603	to choose	1.6 x 0.8	indicative
17	1	LP1 Light pipe PMMA 3 mm	to choose	3 x 3	indicative
18	1	C1 Hold-up supercapacitor	to size	18 x 12	indicative
19	1	TS1 Tamper switch (mounting)	to choose	6 x 6	indicative
20	1	PL1 Tamper plunger, 4 mm, spring loaded	to choose	4 x 4	indicative
21	1	BT1 Li-Po 1S 2000 mAh, protected	Adafruit 2011	60 x 36	distributor
22	1	R1 CC1 pull-down 5.1 kOhm	0402	1 x 0.5	indicative
23	1	R2 CC2 pull-down 5.1 kOhm	0402	1 x 0.5	indicative
24	1	Base with strap ears	RL-HW-01	112 x 62	CAD model
25	1	Lid (non-metallic polymer)	RL-HW-02	92 x 62	CAD model
26	4	Plastic self-tapping screw	to choose	D2.7	to define
27	1	Adhesive foam under the battery	to choose	60 x 36 x 0.5	to define
28	1	Strap or mounting plate	to choose	25 wide	to define

Sources: hardware/netlist.json. "indicative": size to be taken from the part chosen.

- 92 x 62 x 22 mm (112 mm with the strap ears)
- 23 electronic items, one cable inside
- Cellular, satellite and DECT NR+ mesh from one SiP
- Shock +-200 g, motion, tamper plunger, black box flash
- Hold-up capacitor for the last message

GENERAL NOTES

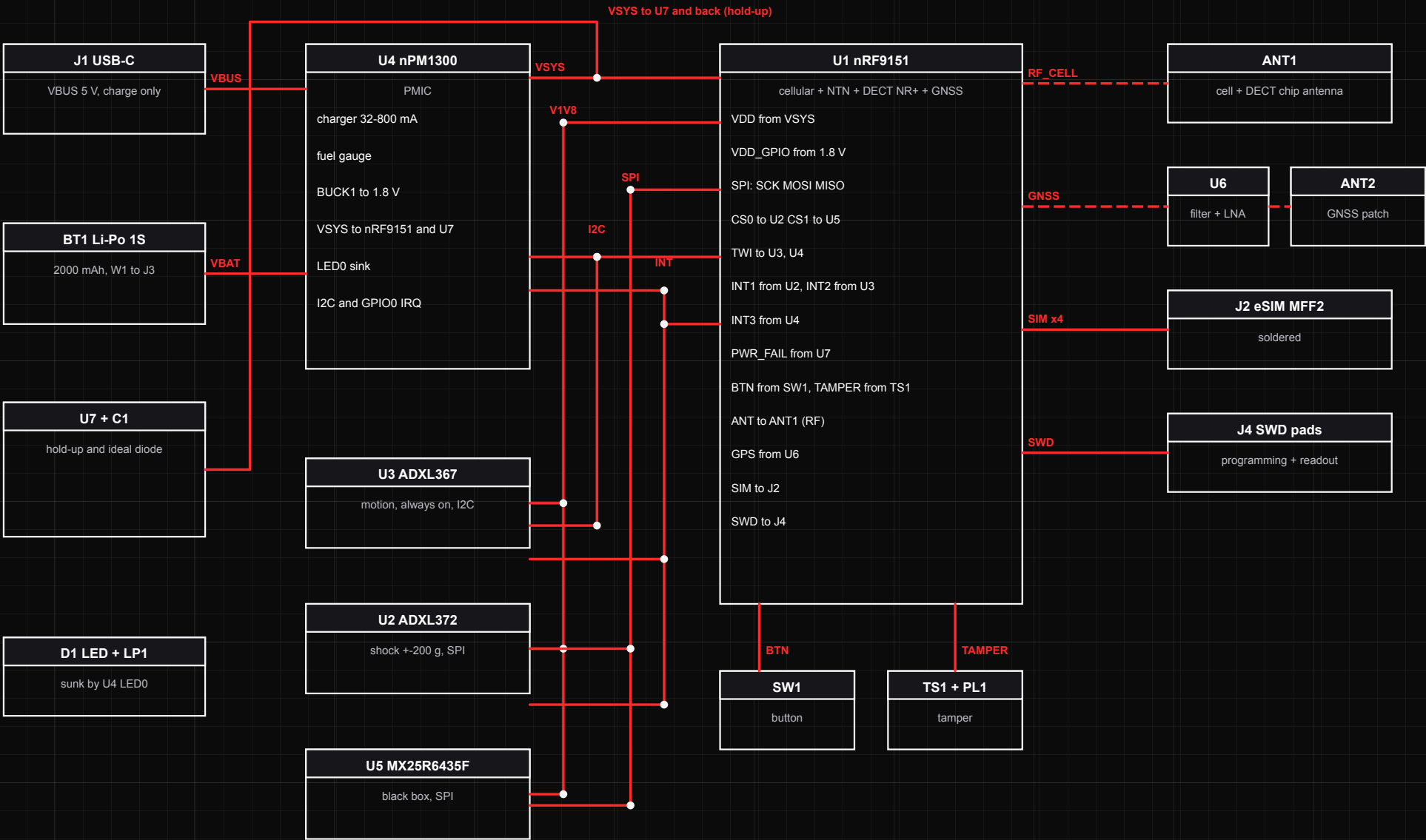
- Dimensions in mm. Third-angle projection. General tolerances to be defined (ISO 2768-m proposed).
- Board outlines: maker or distributor data cited in hardware/netlist.json; others are indicative envelopes.
- Heights, connectors and positions of openings are indicative and must be measured on the chosen parts.
- Geometry and sheets generated by cad/generate.py (build123d 0.13.0); checks run with --check.

EXPLODED VIEW: EVERY COMPONENT

Redline tracker and black box, board RL-MB-01 (nRF9151)

DRAWING NO. RL-HW-03	SCALE 1.15:1 (exploded)	REV 0	SHEET 3 / 5
DRAWN BY: M. ABAAQIL	DATE: 2026-10-05	MATERIAL: to be defined	

BLOCK DIAGRAM FROM hardware/netlist.json



Red solid: signals and supplies. Dashed: 50 ohm RF. One cable inside the device: W1 (BT1 to J3).

Hold-up: C1 0.5 F from 5 V to 3.3 V stores 3.53 J; the last-gasp budget needs 1.50 J (2.4x margin, assumptions in the netlist).

REV	DESCRIPTION	DATE
0	Hardware concept RL-MB-01	2026-10-05

RAILS (RANGES CHECKED)

RAIL	SOURCE	VOLTAGE	LOADS
VBUS	J1 USB-C	4.5-5.5 V	U4
VBAT	BT1 Li-Po 1S	3-4.2 V	U4
VSYS	U4 nPM1300 VSYS	3-5.5 V	U1 VDD, U7, D1
V1V8	U4 nPM1300 BUCK1	1.8 V	U1 GPIO, U2, U3, U5, U6, J4
VCAP	C1 hold-up capacitor	3.3-5 V	C1 via U7

BUSES

BUS	MEMBERS	SELECT / ADDRESS
SPI	U2 ADXL372, U5 flash	CS_IMPACT, CS_BLACKBOX
I2C	U3 ADXL367, U4 nPM1300	0x1D, 0x6B (to confirm)
RF	U1 to ANT1 ; ANT2 to U6 to U1	50 ohm, tuning to do

FIRMWARE POINTS TO FIX

- VDD_GPIO (V1V8) must rise more than 6 ms after VDD and ENABLE: sequence BUCK1 of the nPM1300.
- I2C addresses of the ADXL367 and the nPM1300 are to be confirmed on their data sheets before routing.
- nRF9151 GPIO assignment (CS0, CS1, INT1-3, BTN, TAMPER, PWRFAIL, COEX0) is fixed at routing.
- Black box readout: no USB data on this board. Read the flash through the programming pads (SWD or a UART bridge on them) after recovery, or upload it over LTE-M when a link exists.

COMMUNICATION LAYERS

- Cellular LTE-M / NB-IoT: normal reporting.
 - Fleet relay over DECT NR+: the message hops through any passing fleet vehicle.
 - Satellite NB-NTN: optional, off by default. Needs its own L-band antenna.
- Priority 3 (theft, tamper, power loss) leaves first on whichever layer is up.

GENERAL NOTES

- Dimensions in mm. Third-angle projection. General tolerances to be defined (ISO 2768-m proposed).
- Board outlines: maker or distributor data cited in hardware/netlist.json; others are indicative envelopes.
- Heights, connectors and positions of openings are indicative and must be measured on the chosen parts.
- Geometry and sheets generated by cad/generate.py (build123d 0.13.0); checks run with --check.

ELECTRICAL ARCHITECTURE

Redline tracker and black box, board RL-MB-01 (nRF9151)

DRAWING NO. RL-EL-04	SCALE diagram, not to scale	REV 0	SHEET 4 / 5
DRAWN BY: M. ABAAQIL	DATE: 2026-10-05	MATERIAL: to be defined	

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REDLINE / RL-MB-01

CONCEPT STUDY - PROVISIONAL DIMENSIONS - DO NOT BUILD WITHOUT REVIEW

NET LIST: 33 NETS, ALL TERMINALS UNIQUE (CHECKED)

NET	CLASS	TERMINALS
VBUS	power	J1.VBUS U4.VBUS
VBAT	power	BT1.+ J3.1 U4.VBAT
VSYS	power	U4.VSYS U1.VDD U7.VIN D1.A
VCAP	power	U7.CAP C1.+
V1V8	power	U4.BUCK1 U1.VDD_GPIO U2.VDD U3.VDD U5.VCC U6.VCC J4.VREF
GND	ground	U4.GND U1.GND U2.GND U3.GND U5.GND U6.GND U7.GND C1.- +9
CC1	signal	J1.CC1 R1.1
CC2	signal	J1.CC2 R2.1
SPI_SCK	bus	U1.SPI_SCK U2.SCLK U5.SCLK
SPI_MOSI	bus	U1.SPI_MOSI U2.MOSI U5.SI
SPI_MISO	bus	U1.SPI_MISO U2.MISO U5.SO
CS_IMPACT	bus	U1.CS0 U2.CS
CS_BLACKBOX	bus	U1.CS1 U5.CS
I2C_SDA	bus	U1.TWI_SDA U3.SDA U4.SDA
I2C_SCL	bus	U1.TWI_SCL U3.SCL U4.SCL
INT_IMPACT	signal	U2.INT1 U1.GPIO_INT1
INT_MOTION	signal	U3.INT1 U1.GPIO_INT2
PMIC_INT	signal	U4.GPIO0 U1.GPIO_INT3
PWR_FAIL	signal	U7.PG U1.GPIO_PWRFAIL
RF_CELL	RF	U1.ANT ANT1.FEED
GNSS_ANT	RF	ANT2.FEED U6.IN
GNSS_RF	RF	U6.OUT U1.GPS
LNA_EN	signal	U1.COEX0 U6.EN
SIM_VCC	bus	U1.SIM_VCC J2.VCC
SIM_CLK	bus	U1.SIM_CLK J2.CLK
SIM_IO	bus	U1.SIM_IO J2.IO
SIM_RST	bus	U1.SIM_RST J2.RST
BUTTON	signal	SW1.1 U1.GPIO_BTN
TAMPER	signal	TS1.1 U1.GPIO_TAMPER
LED_SINK	signal	D1.K U4.LED0
SWDIO	bus	U1.SWDIO J4.SWDIO
SWDCLK	bus	U1.SWDCLK J4.SWDCLK
NRESET	signal	U1.RESET J4.RESET

CONNECTOR PIN-OUTS

J1 USB-C	PIN	NET
	CC1	CC1
	CC2	CC2
	GND	GND
	VBUS	VBUS

J2 eSIM MFF2	PIN	NET
	CLK	SIM_CLK
	GND	GND
	IO	SIM_IO
	RST	SIM_RST
	VCC	SIM_VCC

J3 Battery JST-PH 2	PIN	NET
	1	VBAT
	2	GND

J4 SWD pads	PIN	NET
	GND	GND
	RESET	NRESET
	SWDCLK	SWDCLK
	SWDIO	SWDIO
	VREF	V1V8

SW1 Button	PIN	NET
	1	BUTTON
	2	GND

TS1 Tamper switch	PIN	NET
	1	TAMPER
	2	GND

CABLE W1 (ONLY CABLE INSIDE)

BT1 Li-Po

2000 mAh

red: VBAT

black: GND

J3

JST-PH 2 pins, 50 mm. Check polarity before plugging.

POWER-UP SEQUENCE

1. VSYS from U4 (battery or USB-C).

2. VDD of U1 and ENABLE high.

3. BUCK1 starts 1.8 V more than 6 ms later.

4. VDD_GPIO then follows, sensors wake.

5. C1 charges through U7; PWR_FAIL goes high when full.

6. On a supply drop U7 keeps VSYS up and raises PWR_FAIL: last gasp.

PASSIVE PARTS

Decoupling, RF matching, pull-ups and level parts are added at routing; only R1 and R2 (USB-C CC pull-downs) are modelled here because they define the connector.

GENERAL NOTES

1. Dimensions in mm. Third-angle projection. General tolerances to be defined (ISO 2768-m proposed).

2. Board outlines: maker or distributor data cited in hardware/netlist.json; others are indicative envelopes.

3. Heights, connectors and positions of openings are indicative and must be measured on the chosen parts.

4. Geometry and sheets generated by cad/generate.py (build123d 0.13.0); checks run with --check.

WIRING AND NET LIST

Redline tracker and black box, board RL-MB-01 (nRF9151)

DRAWING NO.	SCALE	REV	SHEET
RL-EL-05	tables	0	5 / 5
DRAWN BY: M. ABAAQIL	DATE: 2026-10-05	MATERIAL: to be defined	

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